036015
Finite Elements Method for Engineering Analysis

Homework Assignment 1 - Numerical Analysis

Winter 2027

Instructions

- Electronic submission of **typed** sheets only (PDF format through moodle).
- Submission file must be named: "*name surname.pdf*". No I.D number or homework number is required in the file name.
- Solution must contain all the relevant derivations, explanations, graphs and conclusions.
- All codes used must be added to the submission files.

Problem 1 - Interpolation

Given the function:

$$f(x) = (x + \cos x)^2 \sin x \tag{1}$$

1. Find an interpolation of $f(x)$ in the interval $[0, 0.5]$ of the form:

   $$f(x) = N_i(x) f_i + O(h^4)$$

   where $N_i(x)$ is a row vector of Lagrange polynomials, f_i is a column vector with the values of $f(x)$ at equally spaced nodes and h = 0.5 is the length of the interval. [15pt]

   (a) Explain what is the length of the vectors N_i, f_i that is required to obtain the desired truncation error? [5pt]

2. Explore how the interpolation polynomial from the previous section approximates the function (1). Plot (in Matlab) the interpolation polynomial. Compare the plot of $f(x)$ (1) with its interpolation. Note: In the subsections, do not recompute the interpolation function. [45pt]

   (a) When plotting the comparison, show the plots in the interval $[-0.4, 1]$.

   (b) Show a graph of the **absolute value** of the error in the interval $[-0.15, 0.65]$.

   (c) Discuss briefly (2-5 sentences) the figure produced in 2b. Does it match your expectations? Is there absolute 0 error, and if so why?

   Recommendation: For future assignments it is recommended to write a general function that calculates your $N_i(x)$ vector based on the interval and the truncation error required. Try not to use symbolic terms. The function that needs to be programmed:

   `N=getLagrangePoly(xi)`

   `xi` - vector of nodal position values.

   `N` - $n \times n$ matrix containing the polynomial coefficients of each Lagrange Polynomial, where n is the truncation error.

   You may make changes to this function as you see fit.

Problem 2 - 1D Integration

The error function is defined as follows:

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \tag{2}$$

Write a code (in Matlab) that will receive x and return $\mathrm{erf}(x)$, and use it to plot a figure of $\mathrm{erf}(x)$ in the interval [0, 1]. Use the Gauss quadrature to calculate the integral. Plot solutions for 1 and 2 integration points. You can use the "erf" function in Matlab as benchmark for comparison. [35pt]

Note: You may not use the `quadgk` Matlab function to complete this assignment!

Good luck!